

(355) Python Programming

Description and Eligibility

Evaluate knowledge of working with structured (procedural), object-oriented, and functional programming using the Python programming language. Any postsecondary or secondary division contestant may enter this application event.

Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials
- Computer or laptop/notebook (USB-Type A or adapter required); a full-size keyboard may be used (no printer is needed) and supporting equipment (extension cords, etc.)
- Contestant must use one of these IDEs: IDLE, Visual Studio, Visual Studio Code, PyCharm
- Only pre-written code that is hard copied is allowed.

Competition Notes

- Method of evaluation: application.
- Carry-in and setup of equipment must be done solely by the contestant within the time allotted.
- All student-produced jobs completed on-site are uploaded electronically and are not printed at the national level.
- Business Professionals of America assume no responsibility for hardware/software provided by the contestant.
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area.
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. **Violation of this rule will result in disqualification.**
- At NLC, contestants will take IT Specialist - Python certification as part of the event. Contestants passing this certification will have 50 points added to their event score. A Certiport profile must be created at www.certiport.com before attending NLC. Contestants must include their BPA contestant ID in their Certiport profile when they register. For more information on the exam, visit www.certiport.com. Reference materials are not allowed for certification testing.
- Length of event: no more than ten (10) minutes orientation, no more than ninety (90) minutes testing time, and no more than ten (10) minutes wrap-up.
 - No more than one hundred twenty (120) minutes for certification test.
- Entries vary by state.

Contest Competencies

- Demonstrate understanding of general programming concepts and Python computer language
- Use programming skills for proper development of a Python computer program
- Demonstrate knowledge of Python computer language
- Implement program logic (algorithms, structured design)
- Use structural design techniques and object-oriented concepts
- Create a Python program using calculations, totals, selection, logical operators, classes, sequential file access, I/O operations, loops, methods, arrays, and data structures (linked lists, structures, etc.)